Time Series Seminar Project

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1 Introduction

This report describes the fit of different models to historical currency exchange data. We aim to find whether classical time series models fit better to certain exchange rates than others. We are focused on the longer timeline, where a bigger part of the times series is processed, but we also investigate crucial events in more detail. As one day prediction is not the best way to judge the correctness of our models we cut down the time series to smaller parts here, where timelines before and after the crucial events are included.

We report on the results of an ordinary random walk and try to beat that in different settings. A random walk is defined as simply predicting the last value that the model saw. More advanced models such as Holt's Winters Forecasting, Simple Exponential Smoothing ([GJ85]), ARIMA ([SS24]) and SARIMA are the main approaches we use for Forecasting. The exchange rates used in this study are three, namely: GBP (Great British Pound) per EUR (EURO), USD (United States Dollar) per EUR and USD per GBP. These were picked due to their relevance for the events that we consider and volume in trading (USD: Ranking 1 (44.15%), EUR: 2 (16.14%) and GBP: 4 (6.40%)). The two big events that we focus on in this study are Brexit and the COVID-19 pandemic and smaller events related to the big two are described when analyzed.

The COVID-19 (6.1) pandemic triggered global instability as lock-downs and widespread fear disrupted economic behavior. Due to the uncharacteristic situation, many investors opted for having cash and sold most of their held assets in stocks and bonds. Big and small companies had a bearish [1] view of the market and were looking to cut costs, be that reducing the number of employees or using the situation to get rid of unfavorable contracts. The isolation caused a halt in trading as goods could no longer be transferred by ships due to fear of the infection spreading. Governments took matters into their own hands, with most of them providing the population with stimulus checks that helped the general population and brought balance back to the economy. Brexit (5.1) on the other hand, did **not directly** impact everyone. This is why in our analysis we hope to see patterns or models that predict some rates better than others. Brexit was the withdrawal of the United Kingdom from the European Union. Both parts took measures to overcome the situation as quickly and smoothly as possible, because both would lose parts of direct fundings from each other.

2 Related Work

The benchmark every exchange rate model has to beat is the random walk. Meese and Rogoff ([MR83]) showed that neither structural models nor univariate time series models forecast the dollar against the pound, Deutsche Mark and Yen better than a random walk at horizons of one to twelve months. This result, known as the Meese-Rogoff puzzle, has proved durable up to

[1] Movement described as believing a fall/decline of the market will come.

current date. [Ros13] concludes in her survey that gains over the random walk are rare, small and unstable, and [EW05] show why exchange rates should behave almost like a random walk when their fundamentals are persistent. ARIMA models are the standard univariate alternative. Because log exchange rates are usually integrated of order one, they are fitted to returns, where autocorrelation is weak, so ARIMA forecasts rarely improve much on the random walk, which is itself just an ARIMA(0,1,0). Exponential smoothing plays a similar role. The value of these models lies mainly in providing a transparent baseline and the linear part of hybrid models. While the mean of exchange rate returns is hard to predict, their variance is not. ARCH ([Eng82]) and GARCH ([Bol86]) models capture volatility clustering, and [Hsi89] showed that they explain much of the nonlinear dependence in daily dollar rates. In a widely cited comparison,[HL05] found that no more complex model clearly beat a simple GARCH(1,1) for the DM/USD rate. ARIMA-GARCH combinations therefore improve interval forecasts even when point forecasts barely change. Two-state Markov model were also introduced to try and model the change and long swings of the US dollar. These attempts also showed little result over the plain random walk. Due to the assumptions of linear regression ([WJS+21]) a direct introduction of it is simply viewed as not logical. That is why [DN90] applied non-parametric regression to exchange rate, but again with little result.

Overall, univariate models struggle to beat the random walk in point forecasts but add value for volatility and direction-of-change forecasting, which motivates viewing multiple rates compared at each other as well as the handling of small shocks by more advanced models.

3 Dataset description

We use three daily exchange rate series. The USD per GBP rate (series DEXUSUK) is taken from FRED (Federal Reserve Board, H.10 release) via the R package quantmod; it is a noon buying rate in New York and starts in 1971. The USD per EUR and GBP per EUR rates (series EXR.D.USD.EUR.SP00.A and EXR.D.GBP.EUR.SP00.A) are the ECB euro reference rates, accessed via the R package ecb; they start in 1999. Inverting a series gives the opposite quote, e.g. EUR per USD instead of USD per EUR. We only use data from 2015 to 2020 for the Brexit analysis and from 2019 to May 2023 for the COVID analysis, so the different start dates do not matter. The plots for these Time Series are found in 1.

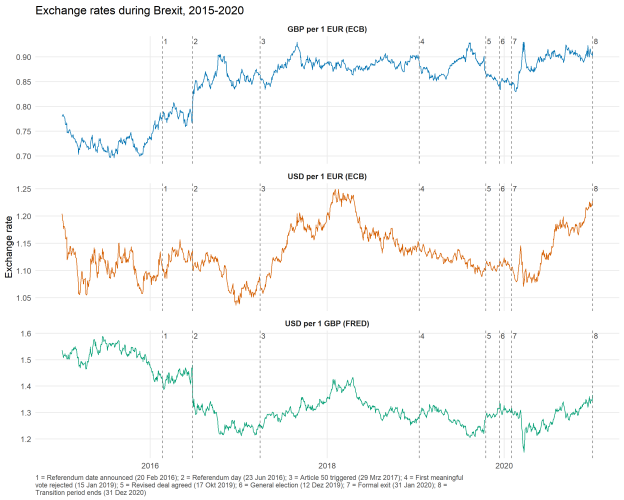


Figure 1: The three time series with the shock events we investiagte in Brexit.

4 Research Questions

In this chapter we introduce the two research questions we investigate in this study. The second research question is investigated only on Brexit events, since with COVID-19 many more single shock events happen close to each other and more aspects of the economy, social and health indexes need to be taken into account. This could be investigated with multivariate time series models, but we leave that for future work.

- **RQ1:** Can univariate time series models capture general patterns in exchange rates and do they forecast better on some rates than others?
- **RQ2:** How well do univariate time series models capture single shock events and does less data hurt the forecasting ability of the model?

5 Brexit Analysis

5.1 Brexit Timeline

David Cameron of the Conservative party of the *United Kingdom*, who served as Prime minister from 2010 to 2016, set Brexit in motion by deciding on a date for the referendum after big discontent in the population. A **European Union (EU)** Referendum Bill was introduced to Parliament on May 28, 2015 and created the legal pathway for the referendum of Stay/Leave for the United Kingdom. By commiting to the referendum on June 23, 2016 the motion for leaving the (EU) became a fact, when 52 % of the voters decided in favor of it. In total, 33.5 million people participated in the referendum, where a lot of disagreement came about. As seen in Figure 2, all subdivisions in *Scotland* voted **against** leaving the EU, while the opposite trend is visible in *England* and *Wales*, with majority of the subdivisions being in favor of the referendum. *Northern Ireland* voters were against leaving the EU. 3 shows how the participants voted by age group. One observes that voting in favor of leaving the EU correlates positively with increasing age. The UK officially left the EU on January 31, 2020 and entered a one year transition period until December 31, 2020. The process was not simple, as the Parliament rejected the proposed withdrawal agreements (by Prime Minister Theresa May) three times. This period was not easy for either of the two sides as many agreements made had to be reviewed. For the populations of both sides this was also not an easy decision to get on board with, as the benefits they had been enjoying for so long like traveling and working would be made more difficult. British companies and the British economy bore most of the economic and social struggle of Brexit. As [2] reports the British economy is smaller than it would have been had it remained in the European Union. This comes as a consequence of difficulties to reach trade agreements for imports and exports, as well as missing the work force needed in sectors where the EU countries could have helped. Trade agreements had to be re-framed and were for the most part advantageous towards the EU, what economist call: likely a penalty for the UK's departure [3]. However, it is not to say that the EU did not struggle from the departure of the United Kingdom as it lost its second biggest economic member. Germany's, France's economic support and a few policies taken by the EU left it struggling less.

[2] https://www.bbc.com/news/articles/cyv0m164m84o

[3] https://www.cer.eu/insights/cost-brexit-ten-years-impact-leaving-customs-union-and-single-market-uk-trade

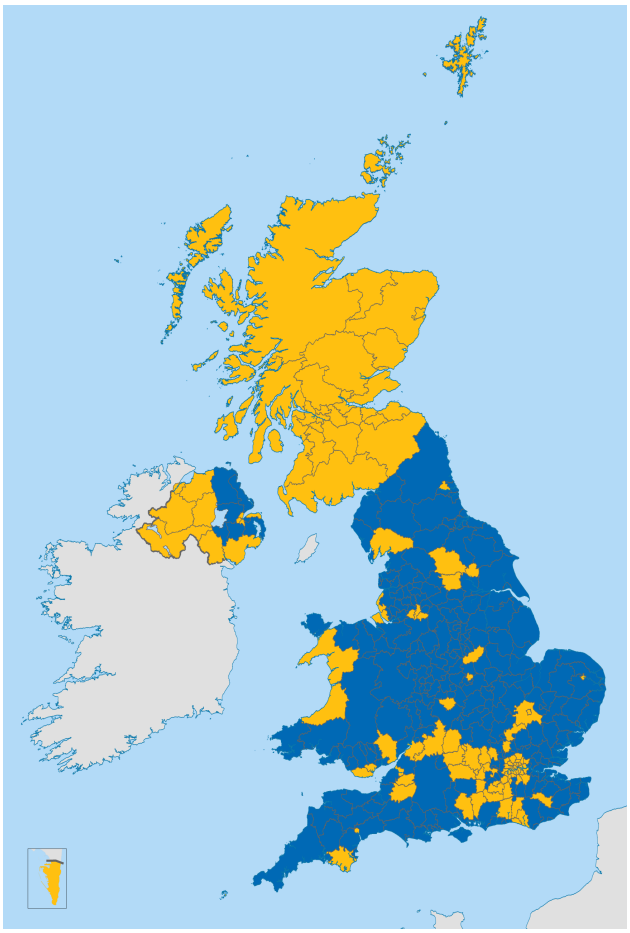


Figure 2: Brexit voting by region, Yellow: Stay, Blue: Leave.

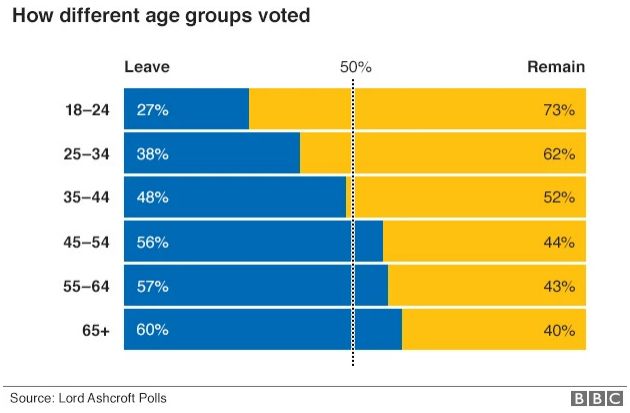


Figure 3: Brexit voting by age, Yellow: Stay, Blue: Leave.

5.2 Time Series Analysis for Brexit

We focus on multiple events and therefore, timelines of Brexit.What interests us are multiple things: Firstly the big picture of how well ARIMA, SARIMA, Holt's Winters Forecasting, Simple Exponential Smoothing forecast the exchange rate and secondly we investigate in closer detail important

events, such as the days/weeks close to the referendum and the days/weeks when the proposals were rejected. The important detail in the second category is the moving average component of the ARIMA and SARIMA. We investigate how well the MA model can handle the shocks to the rates. Table 1 lists the key events that we investigate in detail in this study. The reaction dates are the first trading day on which the markets could react to the event.

ID	Date	Label	Reaction date
1	2016-02-20	Referendum date announced	2016-02-22
2	2016-06-23	Referendum day	2016-06-24
3	2017-03-29	Article 50 triggered	2017-03-29
4	2019-01-15	First meaningful vote rejected	2019-01-16
5	2019-10-17	Revised deal agreed	2019-10-17
6	2019-12-12	General election	2019-12-13
7	2020-01-31	Formal exit	2020-02-03
8	2020-12-31	Transition period ends	2021-01-04

Table 1: Key Brexit Timeline Events

For models that capture seasonality, we only allow for weekly (5 days) seasonality, due to the fact that longer seasonality could backfire when we focus on key events. If model's that capture seasonality do not perform better two things are possible: 1. There is no seasonality, and 2. There is seasonality, but it is not weekly. We leave this for further work to experiment with.

In our setup, we allow 250 days (approximately a year) to firstly be fit for the general Brexit analysis and the next 20 (approximately a month) predicted. In the next step, the training window grows by the 20 data points that were used for forecasting in the past step. Given our period of Brexit is 2015-2020 this would lead to around 60 folds. For our referendum window, we zoom in into ± 3 months and predict only 5 days (one trading week). Again, in the next step the 5 days will be accumulated in the training data. Finally, for key events we train 3 types of models for each of them (Referendum day is repeated here). What differs here is how much past data points are shown into the model. The models are trained on 20, 60 and 120 data points before the key event and are asked to forecast 5 days in three folds, with each fold added 5 days are added to the model. In this setup, we test whether less data points hurt the model's predictive ability for an isolated event or if the model is adaptive enough even on a smaller dataset.

5.3 Results of Time Series Analysis for Brexit

Firstly, we report the results for RQ1 of Brexit with two types of forecasting horizon. The first is the error in 20 days of predicting horizon, while we also care for how the model does after 1, 5, 10 and 20 days as logically when one is forecasting more days the error is likely to grow. The two ways differ in how the averages are taken. In Table 2 we take the average error over the total number of predicted points, while in Table 5 averages are taken over windows sizes 1, 5, 10 and 20. **N** in this table stands for number of folds that were averaged to produce the results. By observing the RMSE ratio (RMSE adv. model/ RMSE naive model) in the whole horizon prediction we conclude that: none of the advanced models do better than the naive approach (**next day = yesterday**). When we focus on different horizons we see that sometimes the naive approach is improved upon, USD/GBP ARIMA and SARIMA with 1 day horizon perform $\sim 4\%$ better than the baseline, however this is not a real result as little trade volatility is moved in 1 day trades. We also see

different pairings beating the baseline by close to 1%, but these models actually fit a SES with α =1 (~ 0.999), which is equal to the naive approach.

For our second approach we present the results of ± 3 months around the referendum period. An interesting fact is that the ARIMA models, when fit on data from 3 months before the referendum and 3 months after the referendum separately, have the same order. The auto.arima call results in ARIMA(0,1,0) otherwise known as a random walk. This shows us that the moving average component, which we were depending on does not matter at all in this scenario. The most significant effect in this analysis is that some of our models beat the naive approach for some periods. When trained and tested on data before the referendum, none of the advanced models perform better than the naive approach ($RMSEratio > 1$). However, when the model has to forecast through the referendum (the referendum falls inside the forecasting folds), we see two models with a $RMSE < 1$. The best is the Holt Winters model on the GBP-EUR pairing, which does include seasonality with a score of 0.94, and the second best is again Holt Winters with a RMSE ratio of 0.964 on the EUR-USD pairing. Thirdly, when the model has seen the referendum in training and forecast beyond it the only model that does better than the naive approach is SARIMA on the pairings EUR-USD with a RMSE score of 0.969 and USD-GBP with RMSE=0.973. However, in Figure 8 we see the general forecasting trends for our models and pairings and deduce that most models fail to capture anything meaningful.

Lastly, for our third approach of focusing on smaller periods of shocking events, with training sets of different sizes we observe: Some events are better predicted than others by more advanced models, however when looking at how well the same model does across all the shocking events the naive approach remains the best. We make this observation after grouping the results by the number of data points that the model saw in training. This again shows that like most things in statistics, when focusing on individual events, human decision might get biased heavily on trusting a single model. In Table 3 we report the best models for our analysis, while in Table 6 we report how many models beat the naive model for each event and amount of lookback.

Pair	**Model**	N	**MSE**	**RMSE**	**MAE**	**MAPE (%)**	**MASE**	**Cov. 95%**	**RMSE ratio**
GBP/EUR	Naive	1229	0.000243	0.015582	0.011138	1.291	3.166	94.22%	1.0000
GBP/EUR	SES	1229	0.000243	0.015582	0.011138	1.291	3.166	94.30%	1.0000
GBP/EUR	Holt	1229	0.000256	0.015998	0.011389	1.323	3.237	93.98%	1.0267
GBP/EUR	Holt-Winters	1229	0.000334	0.018288	0.012811	1.475	3.649	94.63%	1.1737
GBP/EUR	ARIMA	1229	0.000246	0.015678	0.011220	1.302	3.189	94.47%	1.0061
GBP/EUR	SARIMA	1229	0.000244	0.015616	0.011178	1.297	3.177	94.39%	1.0022
EUR/USD	Naive	1229	0.000233	0.015255	0.011533	1.011	2.363	98.70%	1.0000
EUR/USD	SES	1229	0.000233	0.015254	0.011533	1.011	2.363	98.70%	1.0000
EUR/USD	Holt	1229	0.000256	0.015994	0.011889	1.043	2.429	98.45%	1.0485
EUR/USD	Holt-Winters	1229	0.000333	0.018242	0.013722	1.212	2.763	98.78%	1.1959
EUR/USD	ARIMA	1229	0.000237	0.015386	0.011459	1.004	2.351	98.13%	1.0086
EUR/USD	SARIMA	1229	0.000232	0.015240	0.011404	1.000	2.339	98.62%	0.9990
USD/GBP	Naive	1229	0.000640	0.025295	0.017346	1.329	2.727	96.34%	1.0000
USD/GBP	SES	1229	0.000640	0.025295	0.017346	1.329	2.727	96.34%	1.0000
USD/GBP	Holt	1229	0.000660	0.025683	0.017961	1.375	2.824	96.83%	1.0153
USD/GBP	Holt-Winters	1229	0.000833	0.028854	0.020681	1.582	3.237	97.97%	1.1407
USD/GBP	ARIMA	1229	0.000690	0.026271	0.017975	1.377	2.817	96.18%	1.0386
USD/GBP	SARIMA	1229	0.000709	0.026629	0.018658	1.431	2.913	94.79%	1.0527

Table 2: Model Evaluation Metrics Across the Whole Period

Analysis	Pair	Best Model	MASE	MAPE (%)	RMSE vs. Naive
A whole period	GBP-EUR	SES	3.1665	1.2912	1.000001
	EUR-USD	SARIMA	2.3389	0.9996	0.999039
	USD-GBP	Naive	2.7273	1.3295	1.000000
B3 1 before	GBP-EUR	Naive	2.2890	1.0186	1.000000
	EUR-USD	Naive	2.0375	0.6748	1.000000
	USD-GBP	ARIMA	1.8414	0.8581	1.000000
B3 2 contains referendum	GBP-EUR	Holt-Winters	4.8245	2.2566	0.940188
	EUR-USD	Holt-Winters	2.3723	0.8241	0.963722
	USD-GBP	Holt	6.3766	3.2404	0.997723
B3 3 after	GBP-EUR	Holt	1.9815	1.0387	1.005416
	EUR-USD	SARIMA	1.3075	0.4898	0.969530
	USD-GBP	SARIMA	1.5032	1.1108	0.972519
C events, lookback 120	GBP-EUR	Naive	2.4689	1.0471	1.000000
	EUR-USD	Naive	1.7499	0.6268	1.000000
	USD-GBP	Naive	2.4526	1.2177	1.000000
C events, lookback 20	GBP-EUR	Naive	2.3630	1.0471	1.000000
	EUR-USD	Naive	1.9209	0.6268	1.000000
	USD-GBP	Naive	2.1107	1.2177	1.000000
C events, lookback 60	GBP-EUR	Naive	2.4409	1.0471	1.000000
	EUR-USD	Naive	1.8666	0.6268	1.000000
	USD-GBP	Naive	2.3547	1.2177	1.000000

Table 3: Best Performing Models Across Analysis Horizons and Currency Pairs

6 COVID-19 Analysis

6.1 COVID-19 Timeline

COVID-19 was declared a global pandemic on March 11, 2020 by the World Health Organization (WHO), but the general disease and spread of it started on December 12, 2019, thus attaching the number 19 to the name of the virus. The total number of people infected by COVID-19 amounted to a total of 704 million[4] with 7 million of them being fatal. The virus is believed to have been contacted first in China's Hubei Province, in the city of Wuhan. Due to the highly infectious nature of the virus the spread was rapid and worldwide. Countries took severe measures such as lock-downs and isolated treatments for infected individuals. The economic behavior of the general population was out of the ordinary, however, similar to cases of huge economic collapses or catastrophic events. Most people opted to sell their assets, such as stocks, and opted for having cash and safer commodities such as gold and silver.

The market was influenced as goods could not be safely delivered at expected times, but they would take longer periods of time. Oil traded, traded (mostly) in US dollars, hit record time lows and barrel prices dropped below the normal as a consequence of worldwide lockdown. Putting these facts together lead to an abrupt strengthening of the US dollar. However, as things started

[4]https://www.worldometers.info/coronavirus/

to look normal again, the US dollar as most other currencies came back to pre-shock levels and it too got devalued by the during/post pandemic inflation. Inflation worried all socio-economic levels and lead to massive layoffs at small and big companies, making for an unpleasant situation. Many companies believed less production was needed thus less employees were needed. To reverse such trends most countries provided their citizens with stimulus checks and passed new reforms to ease taxes for businesses. Despite these measures [5] (early) reports show multiple countries in the Euro-zone showing declining trends in hiring and purchasing behavior. Following extensive challenges, the World Health Organization officially declared the end of COVID-19 as a Public Health Emergency of International Concern (PHEIC) on May 5,2023. Plots of the time series for COVID-19 are found in Figure 4.

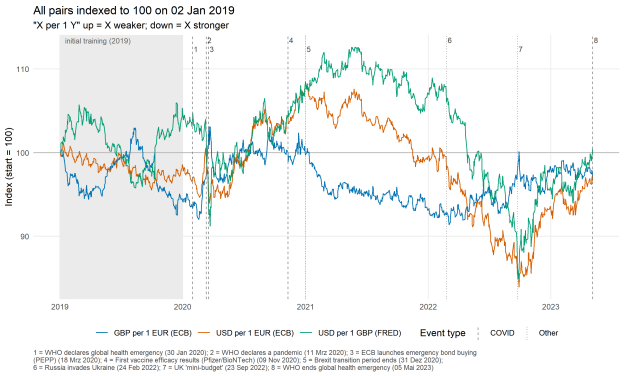


Figure 4: TS for COVID-19.

6.2 Time Series Analysis for COVID-19

In the same setting we only allow for seasonality of 5 days (a week of trading data). The same claim as in the Brexit analysis holds. For COVID-19 we focus our analysis only on RQ1 due to too many single shocks happening during this period.

The results show that even for the COVID data the more advanced models do not capture trends or seasonality better than the random walk. We see SARIMA beating the baseline twice, but that is mostly noise as the difference is less than 1%. What one can focus on is the Coverage that these models achieve. Despite some of them being worse than the baseline we notice that in GBP-EUR Holt and ARIMA reach the same coverage as the baseline. In the pairing EUR-USD Holt-Winters defeats the coverage of the baseline by 1.6% and only ARIMA does worse than the baseline. The issue is that there is no continuation/pattern to these observations, as Holt-Winters performs 2% worse for the rate USD-GBP. Simple Exponential Smoothing should **not** be taken here as a winner method, as similarly to what we found for Brexit α will be set to 1. As reported performance for different horizons in Brexit we do the same in Figure 5. The general trend is the same: The more points the models are asked to forecast, the worse their forecasting gets.

[5] https://www.intereconomics.eu/contents/year/2023/number/6/article/impact-of-the-covid-19-pandemic-on-different-groups-of-smes-in-germany-and-their-recovery.html

Pair	Model	MSE	RMSE	MAE	MAPE	MASE	Coverage 95%	RMSE vs. Naive
GBP-EUR	Naive	0.000156	0.0125	0.0084	0.96%	2.757	96.0%	1.0000
	SES	0.000156	0.0125	0.0084	0.96%	2.758	95.8%	1.0011
	Holt	0.000162	0.0127	0.0085	0.97%	2.789	96.0%	1.0211
	Holt-Winters	0.000238	0.0154	0.0107	1.22%	3.469	95.0%	1.2360
	ARIMA	0.000156	0.0125	0.0084	0.96%	2.757	96.0%	1.0000
	SARIMA	0.000154	0.0124	0.0084	0.96%	2.742	95.9%	0.9943
EUR-USD	Naive	0.000261	0.0161	0.0122	1.10%	3.511	92.7%	1.0000
	SES	0.000261	0.0161	0.0122	1.10%	3.511	92.9%	1.0000
	Holt	0.000273	0.0165	0.0126	1.14%	3.612	92.6%	1.0233
	Holt-Winters	0.000383	0.0196	0.0142	1.27%	4.049	94.3%	1.2123
	ARIMA	0.000263	0.0162	0.0123	1.11%	3.540	91.9%	1.0050
	SARIMA	0.000258	0.0161	0.0121	1.09%	3.465	92.6%	0.9953
USD-GBP	Naive	0.000546	0.0234	0.0172	1.37%	3.099	96.4%	1.0000
	SES	0.000546	0.0234	0.0172	1.37%	3.099	96.4%	1.0000
	Holt	0.000586	0.0242	0.0180	1.43%	3.236	95.9%	1.0362
	Holt-Winters	0.000716	0.0268	0.0195	1.55%	3.491	94.4%	1.1451
	ARIMA	0.000558	0.0236	0.0176	1.39%	3.154	96.4%	1.0111
	SARIMA	0.000554	0.0235	0.0175	1.39%	3.142	96.4%	1.0069

Table 4: Model Performance Metrics for COVID-19 Period Across Currency Pairs

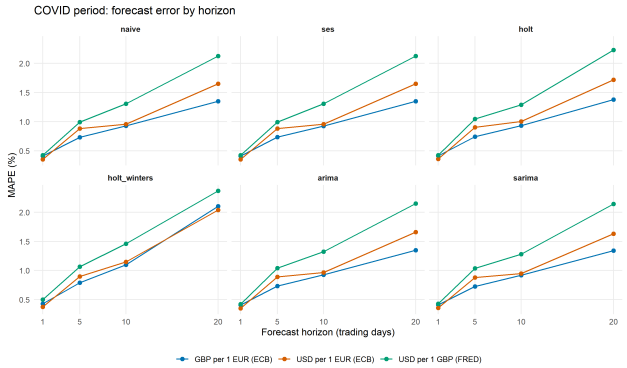


Figure 5: MAPE by horizon for COVID-19.

7 Conclusion

We conclude that no single currency pair is inherently more predictable than another, as all pairs display comparable levels of predictability. Overall, the naive baseline consistently outperforms more complex models. Although certain market shocks and exchange rate behaviors around major Brexit events can be captured by advanced forecasting methods, no single advanced model demonstrates systematic superiority. Furthermore, selecting an advanced approach introduces significant performance variance; a model that forecasts one currency pair effectively is just as likely to yield the poorest predictions for another. Because this model-selection risk cannot be reliably quantified or mitigated, the naive random walk model remains the safest and most dependable forecasting

approach.

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A Performance plots and comparison for Brexit

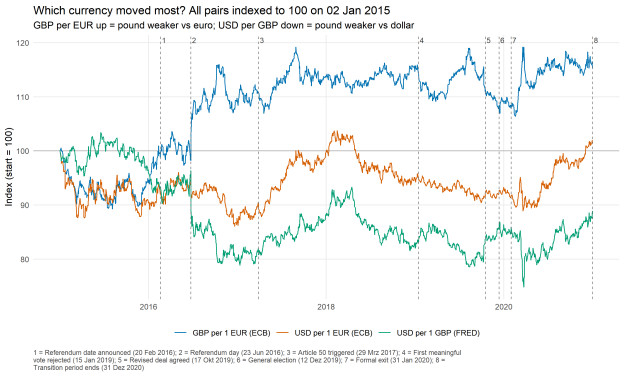


Figure 6: TS indexed at 100 from 2 Jan, 2015. Performance comparison.

In this plot we have indexed the Time Series to have a clearer picture of the rates in terms of percentage. The baseline is 2 January, 2015. We see a decline in the strength of the GBP towards both the Euro and the US dollar. Talks of Brexit and steps taken towards this process seem to have made investors believe more in the Euro and the US dollar. The only series with a clear trend and little shocks is that of GBP per EUR (blue), which shows a total drop of around 15-18% in the strength of the GBP, thus supporting our claim that the UK bore the brunt of the economic consequences of Brexit. Similarly, the US dollar comes out around $\sim 11\%$ stronger compared to the GBP with values falling as far as $\sim 25\%$ lower than the baseline. This would mean that shortly after the formal exit of the UK form European Union one would need only $\frac{3}{4}$'s of the amount of USD in the start of 2020 compared to the 2016 to obtain the same amount of GBP. However, the GBP recovers $\sim 14\%$ in a period of 7-8 months. Finally, the relationship USD-EUR is the most complicated one. At the start it seems that the US dollar is outperforming the Euro, but by the beginning of 2017 and up to early 2018 we see the Euro gaining ground on the USD. This trend is then reversed again, but the relationship seems to then have normalized around a weaker Euro and stronger USD with a difference of around 2-3% in total. Similar trends are observed in 7, when we zoom in on the referendum window.

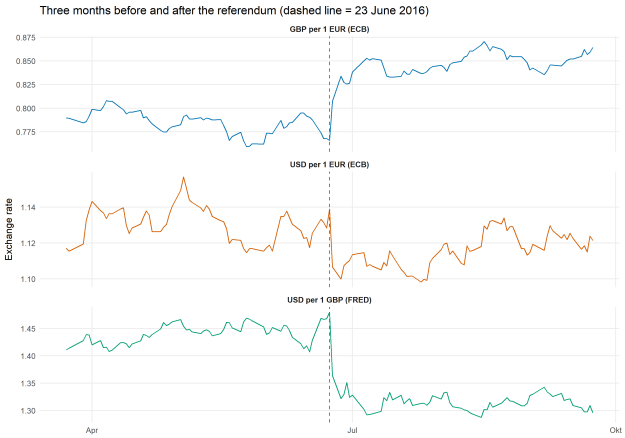


Figure 7: The three rates around the Referendum window.

B Measurements of error for Brexit over different horizon sizes

Table 5: Model Evaluation Metrics by Forecast Horizon

Pair	Model	Horizon	N	MSE	RMSE	MAE	MAPE (%)	MASE	Cov. 95%	RMSE ratio
GBP/EUR	ARIMA	1	62	0.000020	0.004475	0.003250	0.380	0.918	93.55%	0.9980
GBP/EUR	ARIMA	5	62	0.000101	0.010056	0.007423	0.865	2.115	93.55%	1.0025
GBP/EUR	ARIMA	10	61	0.000199	0.014095	0.011110	1.291	3.159	93.44%	0.9997
GBP/EUR	ARIMA	20	61	0.000412	0.020295	0.015068	1.748	4.270	96.72%	1.0099
GBP/EUR	Holt	1	62	0.000020	0.004516	0.003315	0.387	0.935	93.55%	1.0071
GBP/EUR	Holt	5	62	0.000099	0.009934	0.007360	0.858	2.099	93.55%	0.9903
GBP/EUR	Holt	10	61	0.000209	0.014458	0.011367	1.322	3.230	93.44%	1.0255
GBP/EUR	Holt	20	61	0.000444	0.021076	0.015399	1.789	4.362	95.08%	1.0487
GBP/EUR	Holt-Winters	1	62	0.000025	0.005024	0.003879	0.455	1.092	95.16%	1.1204
GBP/EUR	Holt-Winters	5	62	0.000109	0.010459	0.007643	0.884	2.189	96.77%	1.0427
GBP/EUR	Holt-Winters	10	61	0.000245	0.015665	0.012177	1.409	3.464	95.08%	1.1111
GBP/EUR	Holt-Winters	20	61	0.000688	0.026232	0.019637	2.259	5.564	93.44%	1.3053
GBP/EUR	Naive	1	62	0.000020	0.004484	0.003279	0.383	0.925	93.55%	1.0000
GBP/EUR	Naive	5	62	0.000101	0.010031	0.007485	0.872	2.132	93.55%	1.0000
GBP/EUR	Naive	10	61	0.000199	0.014099	0.011048	1.283	3.142	93.44%	1.0000
GBP/EUR	Naive	20	61	0.000404	0.020097	0.014856	1.721	4.211	96.72%	1.0000
GBP/EUR	SARIMA	1	62	0.000020	0.004506	0.003267	0.382	0.923	93.55%	1.0048
GBP/EUR	SARIMA	5	62	0.000099	0.009957	0.007337	0.854	2.090	95.16%	0.9926
GBP/EUR	SARIMA	10	61	0.000196	0.014016	0.011051	1.284	3.142	93.44%	0.9941
GBP/EUR	SARIMA	20	61	0.000410	0.020245	0.015007	1.741	4.252	96.72%	1.0074
GBP/EUR	SES	1	62	0.000020	0.004484	0.003279	0.383	0.925	93.55%	1.0000
GBP/EUR	SES	5	62	0.000101	0.010031	0.007485	0.872	2.132	93.55%	1.0000
GBP/EUR	SES	10	61	0.000199	0.014099	0.011048	1.283	3.142	93.44%	1.0000
GBP/EUR	SES	20	61	0.000404	0.020097	0.014856	1.721	4.211	96.72%	1.0000
EUR/USD	ARIMA	1	62	0.000038	0.006152	0.004517	0.396	0.927	96.77%	0.9944
EUR/USD	ARIMA	5	62	0.000132	0.011472	0.009000	0.791	1.863	98.39%	1.0037
EUR/USD	ARIMA	10	61	0.000268	0.016359	0.011889	1.032	2.444	96.72%	0.9949
EUR/USD	ARIMA	20	61	0.000470	0.021679	0.016969	1.498	3.461	98.36%	1.0324
EUR/USD	Holt	1	62	0.000039	0.006263	0.004601	0.404	0.943	96.77%	1.0124
EUR/USD	Holt	5	62	0.000137	0.011705	0.009113	0.801	1.885	98.39%	1.0241
EUR/USD	Holt	10	61	0.000298	0.017272	0.013095	1.142	2.673	96.72%	1.0504
EUR/USD	Holt	20	61	0.000487	0.022066	0.016917	1.489	3.442	100.00%	1.0508
EUR/USD	Holt-Winters	1	62	0.000041	0.006391	0.004642	0.408	0.949	95.16%	1.0331
EUR/USD	Holt-Winters	5	62	0.000149	0.012197	0.009707	0.856	1.992	100.00%	1.0671
EUR/USD	Holt-Winters	10	61	0.000355	0.018836	0.014659	1.289	2.957	98.36%	1.1455
EUR/USD	Holt-Winters	20	61	0.000631	0.025118	0.019842	1.758	3.985	100.00%	1.1962
EUR/USD	Naive	1	62	0.000038	0.006186	0.004540	0.399	0.932	96.77%	1.0000
EUR/USD	Naive	5	62	0.000131	0.011430	0.008947	0.786	1.852	100.00%	1.0000
EUR/USD	Naive	10	61	0.000270	0.016444	0.012469	1.086	2.558	96.72%	1.0000
EUR/USD	Naive	20	61	0.000441	0.020998	0.016743	1.476	3.414	100.00%	1.0000
EUR/USD	SARIMA	1	62	0.000038	0.006140	0.004522	0.397	0.928	96.77%	0.9926
EUR/USD	SARIMA	5	62	0.000130	0.011422	0.008920	0.784	1.846	100.00%	0.9993

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Table 5 – continued from previous page

Pair	Model	Horizon	N	MSE	RMSE	MAE	MAPE (%)	MASE	Cov. 95%	RMSE ratio
EUR/USD	SARIMA	10	61	0.000262	0.016183	0.011975	1.039	2.464	96.72%	0.9842
EUR/USD	SARIMA	20	61	0.000459	0.021430	0.016844	1.487	3.434	98.36%	1.0206
EUR/USD	SES	1	62	0.000038	0.006187	0.004541	0.399	0.932	96.77%	1.0000
EUR/USD	SES	5	62	0.000131	0.011429	0.008946	0.786	1.852	100.00%	1.0000
EUR/USD	SES	10	61	0.000270	0.016443	0.012468	1.086	2.558	96.72%	1.0000
EUR/USD	SES	20	61	0.000441	0.020998	0.016742	1.476	3.414	100.00%	1.0000
USD/GBP	ARIMA	1	62	0.000076	0.008720	0.006117	0.467	0.959	96.77%	0.9680
USD/GBP	ARIMA	5	62	0.000435	0.020853	0.013393	1.021	2.089	96.77%	1.2088
USD/GBP	ARIMA	10	61	0.000580	0.024085	0.018643	1.412	2.913	95.08%	1.0199
USD/GBP	ARIMA	20	61	0.001242	0.035242	0.024869	1.920	3.883	96.72%	1.0241
USD/GBP	Holt	1	62	0.000082	0.009056	0.006282	0.479	0.982	95.16%	1.0054
USD/GBP	Holt	5	62	0.000307	0.017530	0.012671	0.964	1.995	98.39%	1.0162
USD/GBP	Holt	10	61	0.000593	0.024353	0.018579	1.407	2.909	96.72%	1.0312
USD/GBP	Holt	20	61	0.001236	0.035152	0.025497	1.967	3.984	96.72%	1.0215
USD/GBP	Holt-Winters	1	62	0.000091	0.009533	0.006802	0.519	1.061	95.16%	1.0583
USD/GBP	Holt-Winters	5	62	0.000310	0.017603	0.012930	0.983	2.046	98.39%	1.0204
USD/GBP	Holt-Winters	10	61	0.000681	0.026097	0.020513	1.556	3.196	96.72%	1.1051
USD/GBP	Holt-Winters	20	61	0.001741	0.041722	0.031976	2.461	4.969	98.36%	1.2124
USD/GBP	Naive	1	62	0.000081	0.009008	0.006273	0.478	0.980	95.16%	1.0000
USD/GBP	Naive	5	62	0.000298	0.017251	0.012465	0.949	1.960	98.39%	1.0000
USD/GBP	Naive	10	61	0.000558	0.023616	0.018098	1.371	2.835	95.08%	1.0000
USD/GBP	Naive	20	61	0.001184	0.034413	0.024110	1.862	3.773	96.72%	1.0000
USD/GBP	SARIMA	1	62	0.000076	0.008733	0.005991	0.457	0.938	93.55%	0.9695
USD/GBP	SARIMA	5	62	0.000369	0.019222	0.013841	1.057	2.159	96.77%	1.1142
USD/GBP	SARIMA	10	61	0.000626	0.025020	0.019742	1.497	3.071	95.08%	1.0594
USD/GBP	SARIMA	20	61	0.001294	0.035971	0.025705	1.986	3.998	95.08%	1.0453
USD/GBP	SES	1	62	0.000081	0.009008	0.006273	0.478	0.980	95.16%	1.0000
USD/GBP	SES	5	62	0.000298	0.017251	0.012465	0.949	1.960	98.39%	1.0000
USD/GBP	SES	10	61	0.000558	0.023616	0.018098	1.371	2.835	95.08%	1.0000
USD/GBP	SES	20	61	0.001184	0.034413	0.024110	1.862	3.773	96.72%	1.0000

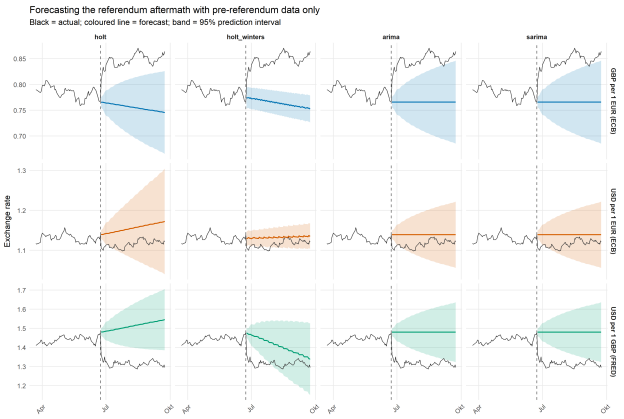


Figure 8: Referendum forecast with models trained on data 3 months before the referendum.

Event	Pair	Best Model	Lookback	Amount of Wins	RMSE vs. Naive
Referendum date announced	Any	Naive	20	0	1
	Any	Naive	60	0	1
	EUR-USD	Holt-Winters	120	2	0.89
Referendum date	EUR-USD	ARIMA	20	9	0.825
	USD-GBP	Holt-Winters	60	1	0.852
	EUR-USD	Holt-Winters	120	1	0.859
Article 50 triggered	GBP-EUR	SARIMA	20	4	0.844
	Any	Naive	60	0	1
	GBP-EUR	Holt	120	1	0.86
First meaningful vote rejected	Any	Naive	20	0	1
	Any	Naive	60	0	1
	Any	Naive	120	1	0.698
Revised deal agreed	Any	Naive	20	0	1
	GBP-EUR	Holt	60	1	0.85
	GBP-EUR	sarima	120	5	0.783
General Election	Any	Naive	20	0	1
	Any	Naive	60	0	1
	Any	Naive	120	0	1
Formal Exit	USD-GBP	Holt-Winters	20	7	0.793
	USD-GBP	Holt-Winters	60	4	0.794
	EUR-USD	Holt	120	3	0.932
Transition Period end	Any	Naive	20	0	1
	USD-GBP	Holt-Winters	60	3	0.794
	USD-GBP	Holt-Winters	120	1	0.903

Table 6: Best Performing Models across shocking Events

Lookback here refers to the amount of data fed to the model during training. Amount of wins is the number of time the more advanced models beat the naive model, here only the best one is reported. The total amount of models for a selected lookback period is 18.

C AI Usage

1. How can I structure the code for a big analysis in R?
2. How can I plot and focus on multiple events in a plot in R?
3. Help me write a function that does ...
4. Check for any grammatical mistakes in this paragraph.
5. What is the best way to structure these two paragraphs?
6. Give me a synonym for "".
7. What is one finding that is considered big in the field of Economics for ARIMA?